EE502 - Linear Systems Theory II **Spring 2032**

Lecture 8

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8.1 Internal (Lyapunov) Stability

In internal stability, we are interested in the un-driven (zero-input response) part of the dynamical system and solely focus on the state evolution dynamics, i.e., the autonomous part of the dynamical system. CT and DT non-linear autonomous systems can simply be expressed as:

$$x \in \mathbb{D} \subset \mathbb{R}^n$$
$$\dot{x} = F(x,t)$$
$$x[k+1] = F(x[k],k)$$

For non-linear systems, in order to define and analyze the stability of a dynamical system, we need to define equilibrium points (or nominal solutions), since we will technically analyze the stability around such points. An equilibrium point for CT and DT non-linear systems is defined as:

$$x_e \in \mathbb{D}$$
$$CT: 0 = F(x_e,t) \ \forall t > t_0$$
$$DT: x_e = F(x_e,k) \ \forall k > k_0$$

Obviously, if a dynamical system at time t_0 (or k_0) starts from an equilibrium point, $x(t_0) = x_e$ (or $x[k_0] = x_e$), it will remain on the equilibrium point $\forall\ t \geq t_0$ (or $\forall\ k \geq k_0$). A non-linear system can have a single equilibrium point, $x_e \in \mathcal{E}$, $\text{card}(\mathcal{E}) = 1$, have a multiple finite number of equilibria, $x_e \in \mathcal{E}$, $\text{card}(\mathcal{E}) = n_e < \infty$, or an infinite number of equilibrium points, $x_e \in \mathcal{E}$, $\text{card}(\mathcal{E}) = \infty$.

Ex 8.1 *Show that for an LTI dynamical system, the set of equilibrium points defines a vector space. Then characterize this vector space.*

Definition: Without loss of generality, let's assume that the equilibrium point that is the point of interest is located at the origin $x_e = 0$.

1. The system is called *stable in the sense of Lyapunov (s.i.s.L)* around $x_e = 0$ if it satisfies:

$$\forall \epsilon > 0,\ \exists \delta_L(\epsilon)\ s.t.\ if\ ||x(t_0)|| < \delta_L\ \rightarrow ||x(t)|| < \epsilon\ \forall t \geq t_0$$

2. The system is called *asymptotically stable* around $x_e = 0$ if it is *stable in the sense of Lyapunov (s.i.s.L)* around $x_e = 0$ and *locally attractive*, i.e.:

$$\exists \delta_a\ s.t.\ if\ ||x(t_0)|| < \delta_a\ \rightarrow \lim_{t\to\infty} ||x(t)|| = 0$$

3. The system is called *exponentially stable* around $x_e = 0$ if it is *asymptotically stable* around $x_e = 0$ and satisfies:

$$\exists \delta_e > 0,\ \alpha > 0,\ \sigma > 0\ s.t.\ if\ ||x(t_0)|| < \delta_e\ \rightarrow ||x(t)|| \leq \alpha ||x(t_0)|| e^{-\sigma t}\ \forall t \geq t_0$$

Remark: If the above stability conditions are satisfied $\forall\, t_0 \in \mathbb{R}$, then we call the system around the equilibrium *uniformly s.i.s.L*, *uniformlyasymptotically stable*, and *uniformly exponentially stable* respectively. The difference between uniform and non-uniform stability is important only for time-varying non-linear systems. Thus, we will not use the uniform stability definition in this course.

Remark: Note that, as you can see, the internal stability definitions, *s.i.s.L*, *asymptotic stability*, and *exponentially stability*, are all local stability definitions defined in the neighborhood of x_e. If the asymptotic and exponential stability definitions hold for all initial conditions, i.e., $x(t_0) \in \mathbb{D}$, then we use the terms *globally asymptotically stable*, and *globally exponentially stable*.

Ex 8.2 *Consider the pendulum dynamics:*

$$\dot{x} = \begin{bmatrix} x_2 \\ -\sin(x_1) \end{bmatrix} \ , \text{ where } \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$$

Analyze the stability of the dynamics around $x_e = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ using Lyapunov's stability definitions.

Solution: To prove stability in the sense of Lyapunov, we need to show that for any given $\epsilon > 0$, we can find a $\delta > 0$ such that if $||x(0)||_2 < \delta$, then $||x(t)||_2 < \epsilon$ for all $t \geq 0$.

We know that a mechanical pendulum is an energy-conserving system. At any time instant, the total energy is:

$$E(x) = \frac{1}{2}x_2^2 + 1 - \cos(x_1) \ , \text{ note } E(0) = 0$$

First, let's find an upper bound for the initial energy $E(x(0))$ purely in terms of the initial state norm $||x(0)||_2$. We use the Taylor series expansion of cosine: $\cos(x_1) = 1 - \frac{x_1^2}{2!} + \frac{x_1^4}{4!} - \dots$. Since this is an alternating series, truncating it after the quadratic term provides an exact algebraic inequality $1 - \cos(x_1) \leq \frac{1}{2}x_1^2$ for all x_1. Thus, the initial energy is bounded by:

$$E(x(0)) = \frac{1}{2}x_2(0)^2 + 1 - \cos(x_1(0)) \leq \frac{1}{2}x_2(0)^2 + \frac{1}{2}x_1(0)^2 = \frac{1}{2}||x(0)||_2^2$$

Second, we find a lower bound for the running energy $E(x(t))$ in terms of $||x(t)||_2$. We can use the exact analytical inequality $1 - \cos(x_1) \geq \frac{2}{\pi^2}x_1^2$, which strictly holds for all $x_1 \in [-\pi, \pi]$. Geometrically, if we plot $1 - \cos(x_1)$ and the parabola $\frac{2}{\pi^2}x_1^2$, both functions equal 0 at the origin and 2 at the boundaries ($\pm\pi$). Because the cosine curve is wider ("bows" out) compared to the specific parabola between these points, $1 - \cos(x_1)$ is always greater than or equal to $\frac{2}{\pi^2}x_1^2$ everywhere in that interval.

Applying this lower bound to the total energy, and noting that $\frac{1}{2} > \frac{2}{\pi^2}$, we get:

$$E(x(t)) = \frac{1}{2}x_2(t)^2 + 1 - \cos(x_1(t)) \geq \frac{1}{2}x_2(t)^2 + \frac{2}{\pi^2}x_1(t)^2 \geq \frac{2}{\pi^2}\left(x_1(t)^2 + x_2(t)^2\right) = \frac{2}{\pi^2}||x(t)||_2^2$$

Now, because the system is energy-conserving, $E(x(t)) = E(x(0))$ for all $t \geq 0$. We can chain our lower and upper bounds together to directly relate $||x(t)||_2$ to $||x(0)||_2$:

$$\begin{aligned} \frac{2}{\pi^2}||x(t)||_2^2 \leq E(x(t)) = E(x(0)) &\leq \frac{1}{2}||x(0)||_2^2 \\ \frac{2}{\pi^2}||x(t)||_2^2 &\leq \frac{1}{2}||x(0)||_2^2 \\ ||x(t)||_2 &\leq \frac{\pi}{2}||x(0)||_2 \end{aligned}$$

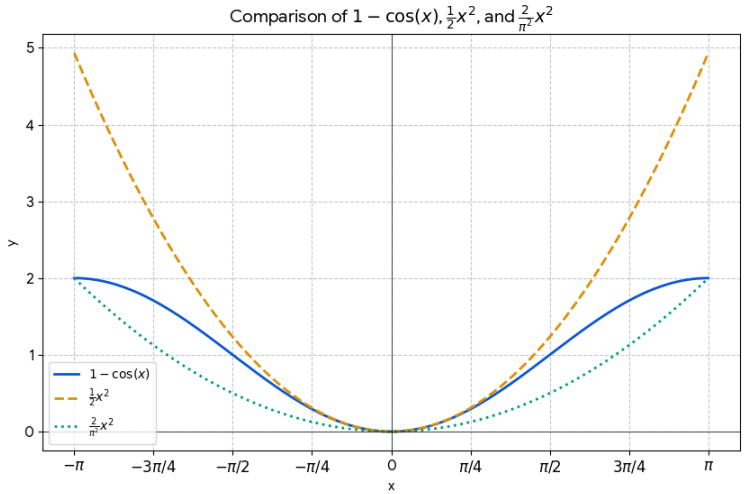

Figure 8.1: Comparison of the potential energy term $1-\cos(x)$ with its quadratic upper bound $\frac{1}{2}x^2$ and lower bound $\frac{2}{\pi^2}x^2$.

This elegant result explicitly shows how the running state is constrained by the initial state! To guarantee the Lyapunov condition $||x(t)||_2 < \epsilon$, we simply require $\frac{\pi}{2}||x(0)||_2 < \epsilon$, which implies $||x(0)||_2 < \frac{2}{\pi}\epsilon$, i.e.

$$\delta(\epsilon) = \min\left(2, \frac{2}{\pi}\epsilon\right)$$

Since we successfully demonstrated that we can find a strictly positive δ for any given $\epsilon > 0$, the dynamics around the equilibrium is *stable in the sense of Lyapunov*. Note that since $E(t) = E(0)\,\forall t > 0$, the state does not converge to the origin over time, meaning the system is not asymptotically stable.

8.1.1 Internal Stability of LTI Systems

Lyapunov's stability definitions may not be very useful for analyzing the stability of non-linear systems; however, we can easily derive necessary and sufficient conditions for stability. One should also note that in LTI systems (and linear systems in general) we are interested in the stability of the origin $x_e = 0$. The following matrix differential and difference equations express autonomous CT and DT LTI systems:

$$\dot{x} = Ax$$
$$x[k+1] = Ax[k]$$

and we know the analytical solutions to the zero-input responses have the following forms:

$$x(t) = e^{At}x_0$$
$$x[k] = A^k x_0$$

It is easier to analyze the internal stability using Jordan decomposition of the system matrix A,

$$x(t) = Ge^{Jt}G^{-1}x_0 \;\rightarrow\; \left[G^{-1}x(t)\right] = e^{Jt}\left[G^{-1}x_0\right] \;\rightarrow\; \alpha(t) = e^{Jt}\alpha_0$$
$$x[k] = GJ^kG^{-1}x_0 \;\rightarrow\; \alpha[k] = J^k\alpha_0$$

Note that since G and G^{-1} are finite and invertible matrices, we know that:

$$\begin{aligned} ||x|| = 0 &\iff ||G^{-1}x|| = 0 \iff x = 0 \\ ||x|| < M_1 < \infty &\iff ||G^{-1}x|| < M_2 < \infty \text{ where} M_1, M_2 \in \mathbb{R} \\ ||x|| \to \infty &\iff ||G^{-1}x|| \to \infty \end{aligned}$$

We know that e^{Jt} and J^k have the following block diagonal form:

$$e^{Jt} = \begin{bmatrix} e^{J_1 t} & & & \\ & e^{J_2 t} & & 0 \\ & & \ddots & \\ & 0 & & \ddots \\ & & & & e^{J_n t} \end{bmatrix}$$

$$J^k = \begin{bmatrix} J_1^k & & & \\ & J_2^k & & 0 \\ & & \ddots & \\ & 0 & & \ddots \\ & & & & J_n^k \end{bmatrix}$$

where an individual block associated with a Jordan block of A has the following form:

$$e^{J_i t} = \begin{bmatrix} e^{\lambda_i t} & te^{\lambda_i t} & \frac{t^2}{2!}e^{\lambda_i t} & \cdots & \frac{t^{n-2}}{(n-2)!}e^{\lambda_i t} & \frac{t^{n-1}}{(n-1)!}e^{\lambda_i t} \\ 0 & e^{\lambda_i t} & te^{\lambda_i t} & \frac{t^2}{2!}e^{\lambda_i t} & \cdots & \frac{t^{n-2}}{(n-2)!}e^{\lambda_i t} \\ \vdots & & \ddots & & & \vdots \\ & & & e^{\lambda_i t} & te^{\lambda_i t} & \frac{t^2}{2!}e^{\lambda_i t} \\ 0 & & \cdots & & e^{\lambda_i t} & te^{\lambda_i t} \\ 0 & & \cdots & & 0 & e^{\lambda_i t} \end{bmatrix}$$

$$J_i^k = \begin{bmatrix} \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} & \binom{k}{2}\lambda_i^{k-2} & \cdots & \binom{k}{n-2}\lambda_i^{k-n+2} & \binom{k}{n-1}\lambda_i^{k-n+1} \\ 0 & \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} & \binom{k}{2}\lambda_i^{k-2} & \cdots & \binom{k}{n-2}\lambda_i^{k-n+2} \\ \vdots & & \ddots & & & \vdots \\ & & & \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} & \binom{k}{2}\lambda_i^{k-2} \\ 0 & & \cdots & & \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} \\ 0 & & \cdots & & 0 & \lambda_i^k \end{bmatrix}$$

Based on this decomposition, we can derive the stability conditions.

1.
   - A CT LTI system is *asymptotically & exponentially stable* $\iff \lim_{t\to\infty} e^{J_i t} = 0 \ \forall i \iff \mathrm{Re}(\lambda_i) < 0 \ \forall i$
   - A DT LTI system is *asymptotically & exponentially stable* $\iff \lim_{k\to\infty} J_i^k = 0 \ \forall i \iff |\lambda_i| < 1 \ \forall i$
2.
   - A CT LTI system is *stable in the sense of Lyapunov (s.i.s.L.)* $\iff \exists M_i < \infty$ s.t. $\|e^{J_i t}\| \le M_i \ \forall t \ge 0, \forall i \iff \big(\mathrm{Re}(\lambda_i) < 0$ or $(\mathrm{Re}(\lambda_i) = 0$ and $\mathrm{size}(J_i) = 1)\big) \ \forall i$

- A DT LTI system is *stable in the sense of Lyapunov (s.i.s.L.)*
  $\iff \exists M_i < \infty$ s.t. $\|J_i^k\| \leq M_i \ \forall k \geq 0, \forall i \iff \big(|\lambda_i| < 1$ or $(|\lambda_i| = 1$ and $\text{size}(J_i) = 1)\big) \ \forall i$

3.
   - A CT LTI system is *unstable*
     $\iff \exists i$ s.t. $\lim_{t\to\infty} \|e^{J_i t}\| = \infty \iff \exists i$ s.t. $\big(\text{Re}(\lambda_i) > 0$ or $(\text{Re}(\lambda_i) = 0$ and $\text{size}(J_i) > 1)\big)$
   - A DT LTI system is *unstable*
     $\iff \exists i$ s.t. $\lim_{k\to\infty} \|J_i^k\| = \infty \iff \exists i$ s.t. $\big(|\lambda_i| > 1$ or $(|\lambda_i| = 1$ and $\text{size}(J_i) > 1)\big)$

8.1.2 Internal Stability of LTV Systems

Indeed, for LTI systems we derived the conditions of stability using the state-transition matrix, which leads us to some conditions based on the eigenvalues of the system matrix. For general LTV systems, if we follow Lyapunov's stability definitions, we can construct the stability conditions.

Theorem: The LTV system with the following state equation and state-transition matrix

$$\dot{x}(t) = A(t)x(t) \ , \ x(t) = \Phi(t, t_0)x_0$$

1. is *s.i.s.L.* $\iff \exists M(t_0)$ *s.t.* $||\Phi(t, t_0)|| \leq M(t_0) < \infty \ , \ \forall t \geq t_0$
2. is *asymptotically stable* $\iff$ if it is *s.i.s.L.* and $\lim_{t\to\infty} \Phi(t, t_0) = 0$
3. is *exponentially stable* $\iff$ if it is *asymptotically stable* and $\exists \sigma(t_0), C(t_0) > 0$ s.t. $||\Phi(t, t_0)|| \leq C(t_0)e^{-\sigma(t_0)(t-t_0)} \ \forall t \geq t_0$

If these stability definitions hold $\forall t_0 \in \mathbb{R}$, then they become *uniformly stable* in their respective categories. Note that the same conditions are valid for the stability of DT LTV systems if we replace time variables.

Ex 8.3 *Let*

$$x[k+1] = \begin{bmatrix} 0 & \rho^k \\ 0 & (1/2) \end{bmatrix} x[k]$$

Analyze the stability of the origin based on the value of ρ.

Solution: In LTI systems, we know that the eigenvalues of the system matrix provide necessary and sufficient conditions for Lyapunov, asymptotic, and exponential stability. We can see that for this LTV system, the eigenvalues are independent of k, and they satisfy LTI stability conditions, i.e.,

$$\lambda(A[k]) \in \{0 \ , \ 1/2\} \& |\lambda_{1,2}| < 1$$

Let's try to find a solution to the initial value problem for the given LTV system and see if the LTI intuition is valid or not for this LTV system. Let $x_{0,1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then

$$x[k]_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

now let $x_{0,2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$x[k]_2 = \begin{bmatrix} (\rho/2)^{k-1} \\ (1/2)^k \end{bmatrix} \quad k > 0$$

thus we can write $\Phi[k, 0]$ as:

$$\Phi[k, 0] = \begin{bmatrix} x[k]_1 & x[k]_2 \end{bmatrix} = \begin{bmatrix} 0 & (\rho/2)^{k-1} \\ 0 & (1/2)^k \end{bmatrix} \quad k > 0$$

Based on this result, we can conclude that:

- The LTV system around the origin is asymptotically & exponentially stable if $|\rho| < 2$
- The LTV system around the origin is stable i.s.L if $|\rho| \leq 2$
- The LTV system around the origin is unstable if $|\rho| > 2$

As you can see in this example (and in general) for LTV systems, the eigenvalues of the system matrix $A[k]$ do not imply stability.

Ex 8.4 *Let*

$$\dot{x} = \begin{bmatrix} 0 & e^{-t} \\ 0 & 0 \end{bmatrix} x$$

Analyze the stability of the origin.

Solution: Since

$$A(t) = \begin{bmatrix} 0 & e^{-t} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} e^{-t} = \bar{A} f(t)$$

We can find the state transition matrix via:

$$\Phi(t, t_0) = e^{\int_{t_0}^{t} A(\tau) d\tau} = \exp\left(\int_{t_0}^{t} \begin{bmatrix} 0 & e^{-\tau} \\ 0 & 0 \end{bmatrix} d\tau \right)$$
$$= \exp\left(\begin{bmatrix} 0 & e^{-t_0} - e^{-t} \\ 0 & 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & e^{-t_0} - e^{-t} \\ 0 & 1 \end{bmatrix}$$

We can see that $||\Phi(t, t_0)|| < M < \infty \ \forall \ (t, t_0)$ but $\lim_{t \to \infty} \Phi(t, t_0) \neq 0$ thus the system around the origin is (only) stable in the sense of Lyapunov. However, if we had only checked the eigenvalues, we may have (wrongly) concluded that the system was unstable.

8.1.3 Lyapunov's Direct Method

Consider an autonomous time-invariant CT or DT non-linear state-space model;

$$\dot{x} = F(x(t))$$
$$x[k+1] = F(x[k])$$

with an equilibrium point x_e. Without loss of generality, let's assume that $x_e = 0$. Consider a continuous scalar function $V : \mathbb{R}^n \mapsto \mathbb{R}$ (or $V : \mathcal{D} \mapsto \mathbb{R}$ in general, where $\mathcal{D}$ is the domain), where $V(0) = 0$ and $V(x) > 0$ for $x \neq 0$ inside some "ball" enclosing the equilibrium point. In that context, $V(x)$ can be considered as an "energy" function (indeed, for a mechanical system, it can be a very good choice), or mathematically $V(x)$ can be a norm. Then $\dot{V}(x)$ for CT systems and $\Delta V(x) := V(x[k+1]) - V(x[k])$ for DT systems can provide critical information regarding the internal stability of the dynamical system.

Lyapunov Functions, Lyapunov Theorem for Local & Global Stability

Definition: Let $V : \mathbb{R}^m \mapsto \mathbb{R}$.

- We call $V(x)$ a *locally positive definite (lpd)* function around $x = 0$ if
  1. $V(0) = 0$
  2. $V(x) > 0,\ ||x|| \in (0, r)$ for some $0 < r$
- We call $V(x)$ a *locally negative definite (lnd)* function around $x = 0$ if
  1. $V(0) = 0$
  2. $V(x) < 0,\ ||x|| \in (0, r)$ for some $0 < r$
- We call $V(x)$ a *locally positive semi-definite (lpsd)* function around $x = 0$ if
  1. $V(0) = 0$
  2. $V(x) \geq 0,\ ||x|| \in (0, r)$ for some $0 < r$
- We call $V(x)$ a *locally negative semi-definite (lnsd)* function around $x = 0$ if
  1. $V(0) = 0$
  2. $V(x) \leq 0,\ ||x|| \in (0, r)$ for some $0 < r$

Definition: Let $V(x)$ be a *lpd* function, then we call it a "candidate" Lyapunov function. If $\dot{V}(x)$ (or $\Delta V(x)$) for the dynamical system of interest is an *lnsd* function, then $V(x)$ is called a **Lyapunov function** of the system.

Theorem: If there exists a Lyapunov function of the system, $V(x)$, then:

1. Then $x_e = 0$ is a (locally) stable equilibrium point *in the sense of Lyapunov.*
2. And if $\dot{V}(x) < 0$ (or $\Delta V(x) < 0$) for $||x|| \in (0, r_a)$ (i.e., *lnd*), then $x_e = 0$ is a (locally) asymptotically stable equilibrium point.

Theorem: If there exists a Lyapunov function of the system, $V(x)$, defined on the entire state-space (or domain in general), i.e.:

$$\exists V : \mathbb{R}^n \to \mathbb{R}\ s.t.\ V(0) = 0\ \ ,\ V(x) > 0,\ \forall x \neq 0\ (i.e.,\ pd) \text{ and}$$
$$\dot{V}(x) \leq 0,\ \forall x \in \mathbb{R}^n\ (i.e.,\ nsd)$$

then:

1. Then $x_e = 0$ is a stable equilibrium point *in the sense of Lyapunov* (in a more global sense).
2. If $x_e = 0$ is a globally stable equilibrium point *in the sense of Lyapunov* and if:

$$\dot{V}(x) < 0,\ \forall x \neq 0$$
$$||x|| \to \infty \implies V(x) \to \infty$$

then $x_e = 0$ is a globally asymptotically stable equilibrium point.

Ex 8.5 *Let's go back to the pendulum dynamics:*

$$\dot{x} = \begin{bmatrix} x_2 \\ -\sin(x_1) \end{bmatrix}\ ,\ \text{where}\ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$$

Analyze the stability of the dynamics around $x_e = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ using Lyapunov's direct method.

Solution: We first need to find a candidate Lyapunov function. Let's test if the total mechanical energy can be adopted as a Lyapunov candidate:

$$\begin{aligned} &V(x) = E(x) = \frac{1}{2}x_2^2 + 1 - \cos x_1 \\ &V(0) = 0 \\ &V(x) > 0 \ , \ \forall x \ s.t. \ \{x \neq 0 \,\&\, x_2 \in (-\pi, \pi)\} \end{aligned}$$

Now let's test if $V(x)$ is a Lyapunov function.

$$\begin{aligned} \frac{d}{dt}V(x) &= -x_2 \sin x_1 + \sin(x_1)x_2 = 0 \\ \dot{V}(x) &\leq 0 \ \forall x \end{aligned}$$

Thus we can conclude that $E(x)$ is a Lyapunov function and $x_e = 0$ is a stable equilibrium point in the sense of Lyapunov.

We already know that since the system conserves its initial energy, thus $\dot{V}(x) = 0 \ \forall x$, $x_e = 0$ is not asymptotically stable. However, one should know the fact that Lyapunov's direct method, in general, does not imply instability.

Ex 8.6 *Now let's add a linear damping term to the pendulum dynamics:*

$$\dot{x} = \begin{bmatrix} x_2 \\ -\sin(x_1) - x_2 \end{bmatrix}$$

Analyze the stability of the dynamics around $x_e = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ using Lyapunov's direct method.

Solution: Total mechanical energy can again be a good candidate as a Lyapunov function:

$$\begin{aligned} &V(x) = E(x) = \frac{1}{2}x_2^2 + 1 - \cos x_1 \\ &V(0) = 0 \ , \ V(x) > 0 \ , \ \forall x \neq 0 \end{aligned}$$

$$\begin{aligned} \frac{d}{dt}V(x) &= -x_2^2 - x_2 \sin x_1 + \sin(x_1)x_2 = -x_2^2 \leq 0 \\ \dot{V}(x) &= 0, \ \forall x \in \{x | \begin{bmatrix} 0 & 1 \end{bmatrix} x = 0\} \\ \dot{V}(x) &< 0, \ \forall x \in \{x | \begin{bmatrix} 0 & 1 \end{bmatrix} x \neq 0\} \end{aligned}$$

In this case we can again conclude that $E(x)$ is a Lyapunov function and $x_e = 0$ is a stable equilibrium point in the sense of Lyapunov. However, we still cannot conclude that the system is asymptotically stable (even though intuitively, we know that it is at least locally asymptotically stable), since $\dot{V}(x)$

is an *nsd* function (locally and globally). Let's try another candidate $V(x)$:

$$V(x) = \frac{1}{2}x_2^2 + a(1-\cos(x_1)) + \frac{b}{2}(x_1+x_2)^2,\ a>0,\ b>0$$
$$V(0) = 0\ ,\ V(x) > 0\ ,\ \forall x \neq 0\ \rightarrow\ pd\ \checkmark$$

$$\begin{aligned}\frac{d}{dt}V(x) &= -x_2^2 - x_2\sin(x_1) + a\sin(x_1)x_2 + b(x_1+x_2)\dot{x}_1 + b(x_1+x_2)\dot{x}_2\\ &= (b-1)x_2^2 + (a-1)x_2\sin(x_1) + bx_1x_2 + b(x_1+x_2)\dot{x}_2\\ &= (b-1)x_2^2 + (a-1)\sin(x_1)x_2 + bx_1x_2 - b\sin(x_1)(x_1+x_2) - bx_2(x_1+x_2)\\ &= -x_2^2 + (a-b-1)\sin(x_1)x_2 - b\sin(x_1)x_1\end{aligned}$$

Let $(a,b) = (2,1)$, then:

$$\frac{d}{dt}V(x) = -\left(x_2^2 + \sin(x_1)x_1\right) \leq 0$$
$$\frac{d}{dt}V(x) = 0 \iff x_1 = k\pi\ k \in \mathbb{Z} \rightarrow\ l.n.d$$

We can now see that when $x_1 \in (-\pi,\pi)$, $\dot{V}(x)$ is a locally negative definite function; the system around $x_e = 0$ is also locally asymptotically stable.

8.1.4 Lyapunov's Indirect Method

Consider an autonomous time-invariant CT or DT non-linear state-space model;

$$\dot{x} = F(x(t))$$
$$x[k+1] = F(x[k])$$

with an equilibrium point x_e. Without loss of generality, let's assume that $x_e = 0$. Let the linearization of the dynamics around the equilibrium yield:

$$\dot{\delta x} = A\delta x(t)$$
$$\delta x[k+1] = A\delta x[k]$$

Theorem:

1. If the linearized system matrix, A, is asymptotically/exponentially stable, then the non-linear system is also *locally asymptotically stable* around the equilibrium point, x_e, inside a region locally defined around the equilibrium point.
2. If the linearized system matrix, A, is asymptotically/exponentially unstable, then the non-linear system is also *locally unstable* around the equilibrium point.
3. If the linearized system matrix, A, is stable i.s.L., we cannot comment on the stability of the non-linear system around the equilibrium.

Ex 8.7 *Consider the following scalar DT dynamical system:*

$$x[k+1] = \frac{x[k]}{1+(x[k])^2}$$

Analyze the stability of the dynamics around $x_e = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ using both Lyapunov's direct and indirect methods.

Solution: Let's first start with the direct method and see if $V(x) = x^2$ is a Lyapunov function:

$$V(x) = x^2 \ , \ V(0) = 0 \ , \ V(x) > 0 \ , \ \forall x \neq 0 \ \rightarrow \ pd \checkmark$$
$$\Delta V(x) = x^2 \left(\frac{1}{(1+x^2)^2} - 1 \right) \ , \ \Delta V(0) = 0 \ , \ \Delta V(x) < 0 \ , \ \forall x \neq 0 \ \rightarrow \ nd \checkmark$$

Thus we can conclude that the system around the origin is globally asymptotically stable. Now let's adopt the indirect method:

$$\delta x[k+1] = \frac{\partial}{\partial x} \left(\frac{x}{1+x^2} \right) \Bigg|_{x=0} \delta x[k]$$
$$\delta x[k+1] = \delta x[k]$$

We can see that the linearized dynamics has a single eigenvalue at $z = 1$, which implies that the linear dynamical system is stable i.s.L. However, based on only this result, we cannot comment on the stability of the non-linear system. However, we know from the direct method that the system around the origin is, indeed, asymptotically stable.

8.1.5 Quadratic Lyapunov Functions and Lyapunov Stability for LTI Systems

Consider the function:

$$V(x) = x^T P x, \ x \in \mathbb{R}^n, P \in \mathbb{R}^{n \times n}, \& \ P^T = P$$

Proposition: $V(x)$ is a positive definite function if and only if all the eigenvalues of P are positive.

Proof: Since P is a symmetric real matrix, we know that it can be diagonalized by an *orthogonal matrix*, i.e.:

$$P = U^T \Lambda U, \ U^T U = I, \& \ \Lambda = \text{diag}\left[\lambda_1, \quad \cdots, \quad \lambda_n\right]$$

Let $y = Ux \ \rightarrow \ ||y||_2 = ||x||_2$, then:

$$V(x) = x^T P x = y^T \Lambda y = \sum_{i=1}^{n} \lambda_i y_i^2$$

Thus:

$$V(x) > 0 \ \forall x \neq 0 \quad \Longleftrightarrow \quad \lambda_i > 0 \ \forall i$$

Definition: A symmetric matrix $P^T = P$:

- is called *positive definite* if it satisfies:

  $$x^T P x > 0 \ \forall x \neq 0$$

  and we use the notation $P > 0$ for such matrices.

- is called *positive semi-definite* if:

  $$x^T P x \geq 0 \ \forall x \neq 0$$

  and we use the notation $P \geq 0$ for such matrices.

Corollaries:

- For a symmetric positive definite or symmetric positive semi-definite matrix:

$$\lambda_{min}(P)||x||_2^2 \leq x^T P x \leq \lambda_{max}(P)||x||_2^2$$

- If $P \geq 0$ but $P \not> 0$, then $\lambda_{min} = 0$.

8.1.6 Quadratic Lyapunov Functions for LTI Systems

Consider defining a Lyapunov function candidate of the form:

$$V(x) = x^T P x \;\; P > 0$$

for an LTI CT or DT system. Then for a CT-LTI system, $\dot{x} = Ax$, we have:

$$\begin{aligned} \dot{V}(x) &= \dot{x}^T P x + x^T P \dot{x} = x^T A^T P x + x^T P A x \\ &= x^T (A^T P + PA) x = -x^T Q x \\ Q &= -(A^T P + PA) \end{aligned}$$

note that $Q^T = Q$. Based on Lyapunov's direct method, we conclude that:

- $\dot{x} = Ax$ is stable i.s.L. if $Q = -(A^T P + PA) \geq 0$ or
- $\dot{x} = Ax$ is asymptotically stable if $Q = -(A^T P + PA) > 0$

Similarly for a DT-LTI system, $x[k+1] = Ax[k]$, we have:

$$\begin{aligned} \Delta V(x) &= x^T A^T P A x - x^T P x \\ &= x^T (A^T P A - P) x = -x^T Q x \\ Q &= -(A^T P A - P) = P - A^T P A \end{aligned}$$

note that again $Q^T = Q$. Based on Lyapunov's direct method, we conclude that:

- $x[k+1] = Ax[k]$ is stable i.s.L. if $Q = P - A^T P A \geq 0$ or
- $x[k+1] = Ax[k]$ is asymptotically stable if $Q = P - A^T P A > 0$

Theorem: Given a CT-LTI dynamic system, $\dot{x} = Ax$, and any $Q > 0$, there exists a definite positive solution P of the Lyapunov equation, $Q = -(A^T P + PA)$, if and only if $\dot{x} = Ax$ is asymptotically stable. The solution P, in this case, is unique.

Proof:

1. For a given $Q > 0$, if $P > 0$ is a solution of $Q = -(A^T P + PA)$, then $V(x) = x^T P x$ is p.d. and $\dot{V}(x)$ is n.d., hence $\dot{x} = Ax$ is globally asymptotically stable.
2. Now let's try to find "a" solution of $P > 0$, given $Q > 0$. Let:

$$P = \int_0^\infty e^{tA^T} Q e^{At} dt$$

P is well defined since all eigenvalues of A are in the Open Left-Half Plane (OLHP), and e^{At} exponentially converges to 0. Now:

$$A^TP + PA = \int_0^\infty A^T e^{tA^T} Q e^{At} dt + \int_0^\infty e^{tA^T} Q e^{At} A dt = \int_0^\infty \left(A^T e^{tA^T} Q e^{At} + e^{tA^T} Q e^{At} A \right) dt$$

$$= \int_0^\infty \frac{d}{dt} \left(e^{tA^T} Q e^{At} \right) dt = \left[e^{tA^T} Q e^{At} \right]_0^\infty = 0 - Q$$

$$A^TP + PA = -Q$$

i.e., P satisfies the Lyapunov equation.

3. Let's show P is symmetric:

$$P^T = \int_0^\infty \left[e^{tA^T} Q e^{At} \right]^T dt = \int_0^\infty \left(e^{At} \right)^T Q^T \left(e^{tA^T} \right)^T dt$$

$$= \int_0^\infty e^{tA^T} Q e^{At} dt = P$$

4. Let's show $P > 0$. Let $x \in \mathbb{R}^n$:

$$x^T P x = \int_0^\infty x^T e^{tA^T} Q e^{At} x dt = \int_0^\infty z(t)^T Q z(t) dt \ , \ z(t) = e^{At} x$$

Note that:

$$z(t) = 0 \iff x = 0 \ , \ \forall t < \infty$$

Since $Q > 0$:

$$z(t)^T Q z(t) = 0 \iff z(t) = 0 \ , \ \forall t$$
$$z(t)^T Q z(t) > 0 \ \forall z(t) \neq 0$$

Thus:

$$x^T P x = \int_0^\infty z(t)^T Q z(t) dt = 0 \iff z(t) = 0 \ \forall t \iff x = 0$$

$$x^T P x = \int_0^\infty z(t)^T Q z(t) dt > 0 \ , \ \forall x \neq 0$$

5. Let's show P is unique. Let's prove this by contradiction. Let's assume that $\exists \bar{P} \neq P$ and $\bar{P}^T = \bar{P}$ such that:

$$A^T \bar{P} + \bar{P} A = -Q \ , \ A^T P + PA = -Q$$

Let $W(t) = e^{tA^T} \bar{P} e^{At}$ and let's analyze the following integral:

$$\int_0^\infty -\dot{W}(t) dt = [-W(t)]_0^\infty = W(0) - W(\infty) = \bar{P} - 0 = \bar{P}$$

$$\int_0^\infty -\dot{W}(t) dt = \int_0^\infty - \left(e^{tA^T} A^T \bar{P} e^{At} + e^{tA^T} \bar{P} A e^{At} \right) dt = \int_0^\infty e^{tA^T} \left(- \left(A^T \bar{P} + \bar{P} A \right) \right) e^{At} dt$$

$$= \int_0^\infty e^{tA^T} Q e^{At} dt = P$$

which is the contradiction to the assumption that $\exists\, \bar{P} \neq P$. This completes the proof.

Theorem: Given a DT-LTI dynamic system, $x[k+1] = Ax[k]$, and any $Q > 0$, there exists a definite positive solution P of the Lyapunov equation, $Q = P - A^T PA$, if and only if $x[k+1] = Ax[k]$ is asymptotically stable. The solution P, in this case, is unique.

Proof:

1. For a given $Q > 0$, if $P > 0$ is a solution of $Q = P - A^T PA$, then $V(x) = x^T Px$ is p.d. and $\Delta V(x)$ is n.d., hence $x[k+1] = Ax[k]$ is globally asymptotically stable.
2. Now let's try to find "a" solution of $P > 0$, given $Q > 0$. Let:

$$P = \sum_{k=0}^{\infty} (A^T)^k Q A^k$$

   P is well defined since all eigenvalues of A are inside the unit circle, and A^k exponentially converges to 0. Now:

$$\begin{aligned} P - Q &= \sum_{k=1}^{\infty} (A^T)^k Q A^k = \sum_{n=0}^{\infty} (A^T)^{n+1} Q A^{n+1} = A^T \left(\sum_{n=0}^{\infty} (A^T)^n Q A^n \right) A \\ P - Q &= A^T PA \\ -Q &= A^T PA - P \end{aligned}$$

   i.e., P satisfies the Lyapunov equation.
3. The remaining parts of the proof are left to the student as a take-home problem.

Discussion: Why use Lyapunov's method for LTI systems?

At this point, a natural question arises: *Why do we need Lyapunov's method for LTI systems when we can simply check the eigenvalues of the A matrix?* While it is true that eigenvalues perfectly determine LTI stability, Quadratic Lyapunov Functions are practically and theoretically indispensable for several key reasons:

- **Numerical Computation:** Finding eigenvalues of very large matrices is computationally expensive and numerically sensitive. Solving the Lyapunov matrix equation ($A^T P + PA = -Q$) is a system of linear equations, which is computationally much faster and more numerically stable.
- **Performance & Optimal Control:** The solution matrix P doesn't just tell us *if* the system is stable, but gives us a measure of *how* stable it is. In classical frequency-domain control, we use phase and gain margins to quantify *relative stability*; in modern state-space control, the relationship between Q and P serves a similar purpose. Furthermore, P allows us to directly compute the "energy" of the transient response. If we define a performance cost as the total accumulated state deviation over time, we can evaluate this infinite integral exactly. Recall that $\dot{V}(x) = -x(t)^T Qx(t)$. By integrating both sides from $t = 0$ to $t = \infty$:

$$\begin{aligned} \int_0^{\infty} x(t)^T Qx(t) dt &= \int_0^{\infty} -\dot{V}(x(t)) dt \\ &= -\Big[V(x(t)) \Big]_0^{\infty} \\ &= -\Big(\lim_{t\to\infty} x(t)^T Px(t) - x(0)^T Px(0) \Big) \end{aligned}$$

Since the system is asymptotically stable, $x(t) \to 0$ as $t \to \infty$. Thus, the integral evaluates perfectly to the initial condition's Lyapunov value:

$$\int_0^\infty x(t)^T Q x(t) dt = x(0)^T P x(0)$$

Because it gives us this total accumulated cost algebraically—without having to simulate the system's differential equations over time—it forms the ultimate theoretical foundation for Optimal Control.

- **Robust and Adaptive Control:** As we saw in Section 8.1.1, eigenvalues fail to guarantee stability for Time-Varying or uncertain systems. Quadratic Lyapunov functions provide a robust framework that extends naturally to LTV systems and systems with bounded uncertainties.
- **Connecting Linear to Non-linear:** By Lyapunov's Indirect Method, if we linearize a non-linear system around an equilibrium, the quadratic Lyapunov function $V(x) = x^T P x$ derived from the linear approximation serves as a valid local Lyapunov function for the full non-linear system. The level sets of this $V(x)$ help us estimate the non-linear *Region of Attraction.*

Ex 8.8 ***Connecting Linear and Non-Linear Analysis.*** *Consider the following 2nd order non-linear dynamical system:*

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -2x_1 - 3x_2 + x_1 x_2 \end{aligned}$$

Analyze the local stability of the origin. First, attempt a naive Lyapunov function $V(x) = x_1^2 + x_2^2$. Then, systematically construct a quadratic Lyapunov function using the linearized system and estimate the region of attraction.

Solution:

Step 1: The Naive Approach Fails. A student might first try the simplest positive definite function: $V(x) = x_1^2 + x_2^2$ (which corresponds to $P = I$). Taking the derivative along the system trajectories:

$$\begin{aligned} \dot{V}(x) &= 2x_1\dot{x}_1 + 2x_2\dot{x}_2 \\ &= 2x_1(x_2) + 2x_2(-2x_1 - 3x_2 + x_1x_2) \\ &= 2x_1x_2 - 4x_1x_2 - 6x_2^2 + 2x_1x_2^2 \\ &= -2x_1x_2 - 6x_2^2 + 2x_1x_2^2 \end{aligned}$$

Is $\dot{V}(x)$ locally negative definite? Let's check a point very close to the origin, for example, $x_1 = -0.1, x_2 = 0.1$:

$$\dot{V}(-0.1, 0.1) = -2(-0.1)(0.1) - 6(0.1)^2 + 2(-0.1)(0.1)^2 = +0.02 - 0.06 - 0.002 = -0.042 < 0$$

Wait, what if we check $x_1 = -1$ and $x_2 = 0.1$:

$$\dot{V}(-1, 0.1) = -2(-1)(0.1) - 6(0.1)^2 + 2(-1)(0.1)^2 = +0.2 - 0.06 - 0.02 = +0.12 > 0$$

Since $\dot{V}(x) > 0$ for points arbitrarily close to the origin (specifically in the upper-left and lower-right quadrants), this naive $V(x)$ is *inconclusive.* It fails to prove stability.

Step 2: Systematic LTI Approach (Indirect Method). Let's approach this systematically. The Jacobian matrix evaluated at $x_e = 0$ is:

$$A = \begin{bmatrix} 0 & 1 \\ -2 + x_2 & -3 + x_1 \end{bmatrix}\Bigg|_{x=0} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

The eigenvalues of A are $\lambda_1 = -1$ and $\lambda_2 = -2$. Since the linear system is strictly stable, Lyapunov theory guarantees a valid quadratic Lyapunov function *must* exist.

We solve the continuous-time Lyapunov equation $A^T P + PA = -Q$. To avoid fractions, let's select $Q = 4I = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$:

$$\begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix}$$

Solving this linear system of equations yields a *non-diagonal* matrix:

$$P = \begin{bmatrix} 5 & 1 \\ 1 & 1 \end{bmatrix}$$

Check positive definiteness: $5 > 0$, and $\det(P) = 5(1) - 1(1) = 4 > 0$. Thus, our systematic Lyapunov function candidate is $V(x) = x^T P x = 5x_1^2 + 2x_1x_2 + x_2^2$.

Step 3: Evaluate on the Full Non-Linear System. We now apply this LTI-derived non-diagonal quadratic function to the original non-linear system. The time derivative $\dot{V}(x)$ is:

$$\begin{aligned}
\dot{V}(x) &= \frac{\partial V}{\partial x_1}\dot{x}_1 + \frac{\partial V}{\partial x_2}\dot{x}_2 \\
&= (10x_1 + 2x_2)(x_2) + (2x_1 + 2x_2)(-2x_1 - 3x_2 + x_1x_2) \\
&= 10x_1x_2 + 2x_2^2 - 4x_1^2 - 6x_1x_2 + 2x_1^2x_2 - 4x_1x_2 - 6x_2^2 + 2x_1x_2^2 \\
&= -4x_1^2 + (10 - 6 - 4)x_1x_2 + (2 - 6)x_2^2 + 2x_1^2x_2 + 2x_1x_2^2 \\
&= -4x_1^2 - 4x_2^2 + 2x_1^2x_2 + 2x_1x_2^2
\end{aligned}$$

Thanks to our carefully computed P matrix, the cross-terms (x_1x_2) perfectly cancelled out! We can elegantly factor this result to analyze its sign:

$$\dot{V}(x) = -2x_1^2(2 - x_2) - 2x_2^2(2 - x_1)$$

For $V(x)$ to be a valid Lyapunov function, we need $\dot{V}(x) < 0$ for all $x \neq 0$. Looking at the factored expression, $\dot{V}(x)$ is strictly negative as long as $(2 - x_2) > 0$ and $(2 - x_1) > 0$. Therefore, $\dot{V}(x)$ is strictly locally negative definite inside the region where $x_1 < 2$ and $x_2 < 2$.

Conclusion & Region of Attraction: This proves local asymptotic stability! It clearly demonstrates the power of the LTI Lyapunov equation: without solving for the "skewed" non-diagonal P matrix, we would have been stuck with the inconclusive naive guess. Furthermore, any level set of $V(x) = c$ that fits entirely inside the domain $\{x \in \mathbb{R}^2 \mid x_1 < 2 \text{ and } x_2 < 2\}$ forms a guaranteed trapping region (Region of Attraction) for the non-linear system.